Solutions to "Algebraic Geometry and Arithmetic Curves"

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1 Some topics in commutative algebra

1.1 Tensor products

Exercise 1.1.1. Let $\{M_i\}_{i\in I}, \{N_j\}_{j\in J}$ be two families of modules over a ring A. Then

$$(\oplus_{i\in I} M_i) \otimes_A (\oplus_{j\in J} N_j) \cong \oplus_{(i,j)\in I\times J}(M_i \otimes_A N_j).$$

Proof. There is a chain of natural isomorphisms:

$$\begin{aligned}
&(\oplus_{i\in I} M_i) \otimes_A (\oplus_{j\in J} N_j)\\
&\cong \oplus_{i\in I}\ (M_i \otimes_A (\oplus_{j\in J} N_j))\\
&\cong \oplus_{j\in J}\ \oplus_{i\in I} (M_i \otimes_A N_j)\\
&\cong \oplus_{(i,j)\in I\times J}\ (M_i \otimes_A N_j).
\end{aligned}$$

□

Exercise 1.1.2. There is a unique A-linear map

$$f : \mathrm{Hom}_A(M, M') \otimes_A \mathrm{Hom}_A(N, N') \to \mathrm{Hom}_A(M \otimes_A N, M' \otimes_A N')$$

such that $f(u \otimes v) = u \otimes v$.

Proof. Consider the map $g : \mathrm{Hom}_A(M, M') \times \mathrm{Hom}_A(N, N') \to \mathrm{Hom}_A(M \otimes_A N, M' \otimes_A N')$ sending (u, v) to $u \otimes v$. It is clearly bilinear, so factors uniquely through $\mathrm{Hom}_A(M, M') \otimes_A \mathrm{Hom}_A(N, N')$. □

Exercise 1.1.3. Let M, N be A-modules and let $i : M' \to M, j : N' \to N$ be submodules of M and N, respectively. Then there is a canonical isomorphism

$$(M/M') \otimes_A (N/N') \cong (M \otimes_A N)/(\mathrm{Im}\, i_N + \mathrm{Im}\, j_N).$$

In particular, $(\mathbf{Z}/n\mathbf{Z}) \otimes_{\mathbf{Z}} (\mathbf{Z}/m\mathbf{Z}) = \mathbf{Z}/l\mathbf{Z}$, where $l = \gcd(m, n)$.

Proof. Observe that

$$(M \otimes_A N)/(\mathrm{Im}\, i_N + \mathrm{Im}\, j_N) \cong ((M \otimes_A N)/(\mathrm{Im}\, i_N))/(\mathrm{Im}\, j_{M/M'}),$$

so Corollary 1.13 gives us a chain of isomorphisms

$$\begin{aligned} &((M \otimes_A N)/(\mathrm{Im}\, i_N))/(\mathrm{Im}\, j_{M/M'}) \\ \cong &((M/M') \otimes_A N)/(\mathrm{Im}\, j_{M/M'}) \\ \cong &(M/M') \otimes_A (N/N'). \end{aligned}$$

For the last statement, just recall that $(m) + (n) = (l)$. □

Exercise 1.1.4. Let M, N be A-modules and let B, C be A-algebras.

(a) If M and N are finitely generated over A, then so is $M \otimes_A N$.

(b) If B and C are finitely generated over A, then so is $B \otimes_A C$.

(c) The converses of (a) and (b) are in general false.

Proof. Parts (a) and (b) are obvious. For part (c), just notice that $\mathbf{Q} \otimes_{\mathbf{Z}} \mathbf{Z}/2\mathbf{Z}$ is the zero ring, but $\mathbf{Q}$ is not finitely generated over $\mathbf{Z}$. □

Exercise 1.1.5. Let $\rho : A \to B$ be a ring homomorphism, M an A-module, and N a B-module. There is a canonical isomorphism of A-modules

$$\mathrm{Hom}_A(M, \rho_* N) \cong \mathrm{Hom}_B(\rho^* M, N).$$

Proof. Let $f : \mathrm{Hom}_A(M, \rho_* N) \to \mathrm{Hom}_B(\rho^* M, N)$ be the map sending u to the map $f(u)$ s.t. $f(u)(m \otimes b) = bu(m)$. This is well-defined because if $n \otimes c = m \otimes b$, we may assume that there is some $a \in A$ s.t. $an = m$ and $\rho(a)b = c$ and so $f(u)(n \otimes c) = cu(n) = \rho(a)bu(n) = bu(m)$.

In addition, define $g : \mathrm{Hom}_B(\rho^* M, N) \to \mathrm{Hom}_A(M, \rho_* N)$ to be the map s.t. $g(u)(m) = u(m \otimes 1)$. This is immediately seen to be well-defined, and we contest that this is an inverse for f. Indeed, $f(g(u))(m \otimes b) = bg(u)(m) = bu(m \otimes 1) = u(m \otimes b)$ and $g(f(u))(m) = f(u)(m \otimes 1) = u(m)$. □

Exercise 1.1.6. Let $(N_i)_{i \in I}$ be a direct system of modules. Then for any A-module M, there exists a canonical isomorphism $\varinjlim (N_i \otimes_A M) \cong (\varinjlim N_i) \otimes_A M$.

Proof. Tensor-hom.

Just kidding, it follows from the fact that $\varinjlim (N_i \times M)$ is canonically isomorphic to $(\varinjlim N_i) \times M$, and that a bilinear map over the latter must then induce a cocone of bilinear maps. □

Exercise 1.1.7. Let B be an A-algebra, and let M, N be B-modules. Then there exists a canonical surjection $M \otimes_A N \to M \otimes_B N$.

Proof. Do I really need to spell it out? □

1.2 Flatness

Exercise 1.2.1. Let M be an A-module and let $I \subseteq \mathrm{Ann}(m)$ be an ideal.

(a) There is a natural A/I-module structure on M, and $M \cong M \otimes_A A/I$.

(b) Let N be another A-module such that $I \subseteq \mathrm{Ann}(N)$. The canonical homomorphism $M \otimes_A N \to M \otimes_{A/I} N$ is an isomorphism.

Proof. For (a), observe that if $x \in I$, then $am = (a + x)m$ for any $a \in A$ and $m \in M$. This implies that the multiplication of elements of M by cosets $a + I$ is well defined, hence we have an A/I-module structure on M. To see that this is the extension of scalars, recall that $M \otimes_A A/I \cong M/(IM)$, which is just M.

Part (b) follows from a chain of isomorphisms

$$\begin{aligned} & M \otimes_A N \\ \cong & (M \otimes_A A/I) \otimes_{A/I} N \\ \cong & M \otimes_{A/I} N, \end{aligned}$$

where the first isomorphism comes from Corollary 1.11 and the second comes from part (a). □

1.3 Formal completion

2 General properties of schemes

2.1 Spectrum of a ring

Exercise 2.1.1. For k a field, $\mathrm{Spec}(k[[T]])$ is the space $\{\xi, 0\}$ with generic point ξ and closed point 0.

Proof. It is easy to see that any power series with a non-zero constant term is a unit. Furthermore, if we have $f = \sum a_n T^n$ with $a_N = 1$ for some N and $a_n = 0$ whenever $n \leq N$, then we can pull out a common factor of T^N. We see that f is then a unit multiple of T^N and $(f) = (T^N)$. Clearly, this is only prime when $N = 1$. It is easy to see that $k[[T]]$ is an entire ring, so the proposition is immediate. □

Exercise 2.1.2. Let $\varphi : A \to B$ be a homomorphism of finitely generated algebras over a field k. The image of a closed point under $\mathrm{Spec}\,\varphi$ is a closed point.

Proof. If $\mathrm{Spec}\,\varphi$ takes the maximal ideal $\mathfrak{m}$ into the ideal $\mathfrak{p}$, we get a diagram

$$k \to A/\mathfrak{p} \xrightarrow{\widetilde{\varphi}} B/\mathfrak{m}.$$

By Corollary 1.12, $B/\mathfrak{m}$ is a finite algebraic extension of k, so $B/\mathfrak{m}$ (resp. $A/\mathfrak{p}$) is finite over $A/\mathfrak{p}$ (resp. k). It then suffices to prove that $A/\mathfrak{p}$ is a field. Any $a \in A$

has an associated irreducible polynomial $P = \sum a_i X^i \in k[x]$ such that $P(a) = 0$. This polynomial must have a nonzero constant term, so we have

$$-\frac{1}{a_0} \sum_{i>0} a_i a^i = 1,$$

but the polynomial on the left has no constant term and we can pull out a factor of a. $\square$